

MOHAMED ELSAID

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PROFESSIONAL SUMMARY

Bioinformatics scientist with an M.S. in Bioinformatics from Johns Hopkins University (completed May 2026) and 3 years of pharmaceutical industry experience at Pfizer and Catalent in GMP and GLP regulated labs. Built six complete, version tagged projects on real public data spanning RNA-seq, GATK4 variant calling, 16S microbiome, multi-omics integration, ML, and SQL. Brings regulated data integrity practice (ALCOA+, 21 CFR Part 11, LIMS) to reproducible computational work. Seeking Bioinformatics Analyst, Bioinformatics Scientist, or Computational Biologist roles.

TECHNICAL SKILLS

NGS and Pipelines: Snakemake, GATK4, BWA-MEM, STAR, SAMtools, Trimmomatic, featureCounts, FastQC, MultiQC

Statistical Genomics: DESeq2, clusterProfiler (GSEA), DADA2, phyloseq, vegan, ANCOM-BC2, mixOmics (DIABLO)

Machine Learning: scikit-learn (Logistic Regression, Random Forest, SVM), cross validation, pandas

Programming: Python, R, SQL/SQLite, Bash, Linux, Git/GitHub, Conda, Docker

Regulated Lab: GMP, GLP, LIMS, Trackwise, ALCOA+, 21 CFR Part 11, environmental monitoring, Hamilton liquid handling

BIOINFORMATICS PORTFOLIO PROJECTS

Six complete projects, tagged v1.0, public at github.com/MohamedElsaid-bit

- **RNA-seq Differential Expression Pipeline** (*Snakemake, STAR, DESeq2*): Built FastQC, Trimmomatic, STAR, featureCounts, DESeq2, clusterProfiler GSEA, and MultiQC workflow; ran end to end on public airway dexamethasone data (GEO GSE52778), finding 108 differentially expressed genes (61 up, 47 down) and recovering known response genes DUSP1, PER1, and FKBP5.
- **Germline Variant Calling Pipeline** (*GATK4, Snakemake*): Implemented GATK4 best practices (BWA-MEM, MarkDuplicates, BQSR, HaplotypeCaller, hard filtering) on the real chr21 reference with simulated reads; 71 PASS variants from 89 candidates (68 SNPs, 3 indels), with simulation artifacts in Ti/Tv and dbSNP concordance reported and explained.
- **Gut Microbiome Diversity Analysis** (*DADA2, phyloseq, vegan*): Inferred ASVs and SILVA taxonomy for 19 mouse fecal 16S rRNA samples (early vs late post weaning); found a significant composition shift (PERMANOVA, $p = 0.001$) with no significant change in overall diversity, and 9 genera differentially abundant by ANCOM-BC2.
- **Multi-Omics Integration Capstone** (*mixOmics DIABLO*): Integrated matched mRNA, miRNA, and protein data from the same TCGA breast cancer patients to predict molecular subtype; reported that the best single layer (mRNA, 98.6% test accuracy) outperformed the 3 block integrated model (94.3%).
- **Biomedical ML Classification** (*scikit-learn*): Classified 5 cancer types from TCGA PanCancer RNA-seq (801 tumor samples, 20,531 gene features) at 99.4% test accuracy; Wisconsin Breast Cancer stage (569 samples) reached 98.3% accuracy and 0.995 ROC AUC with stratified 5 fold cross validation and leakage safe Pipelines.
- **Portfolio Analytics** (*SQL, SQLite*): Designed a LIMS style database loaded from the committed outputs of the other five pipelines; wrote 8 queries (joins, window functions, conditional aggregation pivots) and a reporting layer with zero external dependencies.

PROFESSIONAL EXPERIENCE

Associate Scientist, QC Microbiology | Catalent Pharma Solutions, Harmans, MD

Apr 2025 to Oct 2025

- Performed environmental monitoring in ISO 5, 7, and 8 cleanrooms (air, surface, and nonviable particulate sampling) and tested facility utilities including USP purified water and clean steam.
- Ran routine and nonroutine sterility, endotoxin, and bioburden assays under GMP; operated, maintained, and calibrated lab equipment to 5S standards.
- Recorded results in LIMS per SOPs and self reviewed analytical data; supported out of specification (OOS) investigations, tracked events in Trackwise, and assisted with client audits and regulatory inspections.

Associate Scientist | Pfizer Inc., Pearl River, NY

Aug 2022 to Feb 2024

- Received and logged SARS-CoV-2 vaccine clinical trial samples into an electronic sample management database, handling 800 to 1,000 samples per week over 18 months.
- Managed sample storage and retrieval across manual freezers and the BiOS automated freezer storage system; performed manual aliquoting and automated aliquoting on Hamilton robotic instruments.
- Shipped samples and supplies to internal and external testing labs, resolved shipping discrepancies, maintained Hamilton instruments, and handled sample destruction per SOPs under GxP, cGMP, and GLP.

EDUCATION

M.S. in Bioinformatics | Johns Hopkins University, Baltimore, MD

Completed May 2026

B.S. in Biochemistry | Manhattan College, Bronx, NY

May 2022